

1

Agent Tasks

Multi-step agent tasks
with different
workloads and workflows



Forward

Agent Workloads



Coding



SQL



Web
Search



Bash

...

Agent Workflows



ReAct



Plan-Solve



AgentFlow

...

2

Task Scheduling



Feature Extraction

- Workload, Workflow
- Current step
- Prompt keywords



History-based Estimator

- Record running history
- Predicted remaining steps
- Estimated remaining time



Priority Scheduler

- Shorter remaining time first

4

Inference Engine

Non-SD Mode

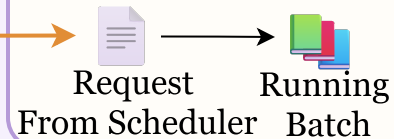
SD Mode

CUDA Graph A

Autoregressive Generation



Continuous Batching



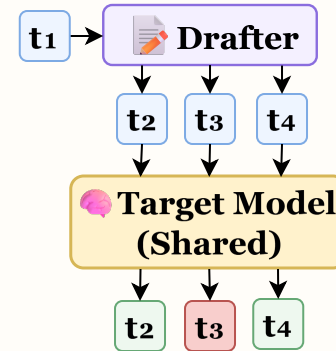
Metrics

Metrics

Switch

CUDA Graph B

Draft + Parallel Verify



3

Adaptive SD Control



Offline Profiler

- Measure SD speedup
- Find beneficial BS range



Online Load Monitor

- Running requests BS
- Resume requests
- Waiting Queue Length



SD Switch Controller

- Enable SD when beneficial
- Disable SD under high load